Problem 1

$$E_x = A e^{ikx-i\omega t} \begin{cases} \exp\left(-z\sqrt{k^2-\omega^2/c^2}\right), & z>0 \\ \exp\left(z\sqrt{k^2-\frac{\varepsilon\omega^2}{\varepsilon_0 c^2}}\right), & z<0 \end{cases} \qquad (1)$$

Similarly, E_z is given in the class notes:

$$E_z = ikAe^{ikx-i\omega t} \begin{cases} \dfrac{\exp\left(-z\sqrt{k^2-\omega^2/c^2}\right)}{\sqrt{k^2-\omega^2/c^2}}, & z>0 \\ -\dfrac{\exp\left(z\sqrt{k^2-\varepsilon\omega^2/\varepsilon_0 c^2}\right)}{\sqrt{k^2-\varepsilon\omega^2/\varepsilon_0 c^2}}, & z<0 \end{cases} \qquad (2)$$

Magnetic field is given by

$$\vec{B} = -\frac{i}{\omega}\nabla\times\vec{E}$$

$$B_x = -\frac{i}{\omega}\left(\frac{\partial}{\partial y}E_z - \frac{\partial E_y}{\partial z}\right) = 0$$

(since $E_y=0$ and E_x, E_z are independent of y)

$$B_z = -\frac{i}{\omega}\left(\frac{\partial}{\partial x}E_y - \frac{\partial E_x}{\partial y}\right) = 0$$

$$B_y = -\frac{i}{\omega}\left(\frac{\partial E_x}{\partial z} - \frac{\partial E_z}{\partial x}\right) = -\frac{i}{\omega}\left(\frac{\partial E_x}{\partial z} - ikE_z\right) \qquad (3)$$

We now substitute Eqs. (1) and (2) into Eq. (3):

for $z>0$:

$$B_y = -\frac{i}{\omega}\left(-\sqrt{k^2-\omega^2/c^2}\,Ae^{ikx-i\omega t}\exp\left(-z\sqrt{k^2-\omega^2/c^2}\right) + k^2Ae^{ikx-i\omega t}\frac{\exp\left(-z\sqrt{k^2-\omega^2/c^2}\right)}{\sqrt{k^2-\omega^2/c^2}}\right) = \qquad (4)$$

$$= -\frac{i\omega}{c^2} A e^{ikx-i\omega t} \frac{\exp\left(-z\sqrt{k^2-\omega^2/c^2}\right)}{\sqrt{k^2-\omega^2/c^2}}$$

for $z<0$:

$$B_y = -\frac{i}{\omega}\left(\sqrt{k^2-\frac{\varepsilon\omega^2}{\varepsilon_0 c^2}}\, A e^{ikx-i\omega t} \exp\left(z\sqrt{k^2-\frac{\varepsilon\omega^2}{\varepsilon_0 c^2}}\right)\right.$$

$$\left. - k^2 A e^{ikx-i\omega t} \frac{\exp\left(z\sqrt{k^2-\frac{\varepsilon\omega^2}{\varepsilon_0 c^2}}\right)}{\sqrt{k^2-\frac{\varepsilon\omega^2}{\varepsilon_0 c^2}}}\right) \qquad (5)$$

$$= \frac{i\varepsilon\omega}{\varepsilon_0 c^2} A e^{ikx-i\omega t} \frac{\exp\left(z\sqrt{k^2-\frac{\varepsilon\omega^2}{\varepsilon_0 c^2}}\right)}{\sqrt{k^2-\frac{\varepsilon\omega^2}{\varepsilon_0 c^2}}}$$

By virtue of the dispersion equation for the surface plasmons:

$$\frac{\varepsilon}{\varepsilon_0} = -\frac{\sqrt{k^2-\frac{\omega^2\varepsilon}{c^2\varepsilon_0}}}{\sqrt{k^2-\frac{\omega^2}{c^2}}}$$

the two expressions (4) and (5) match at $z=0$.

Thus, $\vec{B}$ is indeed continuous at the surface.

Problem 2

$$E_y = E_y(z)\, e^{ikx - i\omega t}$$

Maxwell equation (3) of the notes for E_y is completely analogous to the one for E_x with the same boundary condition (continuity at $z=0$).

Thus

$$E_y = A e^{ikx - i\omega t} \begin{cases} \exp\left(-z\sqrt{k^2 - \frac{\omega^2}{c^2}}\right), & z>0 \\ \exp\left(z\sqrt{k^2 - \frac{\varepsilon\omega^2}{\varepsilon_0 c^2}}\right), & z<0 \end{cases}$$

Since $E_x = 0$ and $E_z = 0$ ($D_z = 0$) <u>all</u> relevant components of $\vec{E}$ and $\vec{D}$ are properly continuous at $z=0$. Let us check the continuity of $\vec{B}$ though:

$$B_x = -\frac{i}{\omega}(\nabla \times \vec{E})_x = -\frac{i}{\omega}\left(\frac{\partial E_z}{\partial y} - \frac{\partial E_y}{\partial z}\right) = \frac{i}{\omega}\frac{\partial E_y}{\partial z}$$

$$B_x = \frac{iA}{\omega} e^{ikx - i\omega t} \begin{cases} -\sqrt{k^2 - \frac{\omega^2}{c^2}}\, \exp\left(-z\sqrt{k^2 - \frac{\omega^2}{c^2}}\right), & z>0 \\ \sqrt{k^2 - \frac{\varepsilon\omega^2}{\varepsilon_0 c^2}}\, \exp\left(z\sqrt{k^2 - \frac{\omega^2}{c^2}}\right), & z<0 \end{cases}$$

This would require that

$$\sqrt{k^2 - \frac{\omega^2}{c^2}} = -\sqrt{k^2 - \frac{\varepsilon\omega^2}{\varepsilon_0 c^2}}$$

Such a solution is impossible (at least when $\sqrt{k^2 - \omega^2/c^2}$ and $\sqrt{k^2 - \frac{\varepsilon\omega^2}{\varepsilon_0 c^2}}$ are real). $\longrightarrow$ no surface TE-wave.